Portfolio Allocation Engine

Interactive Asset Analysis & Portfolio Optimization – Project Goal

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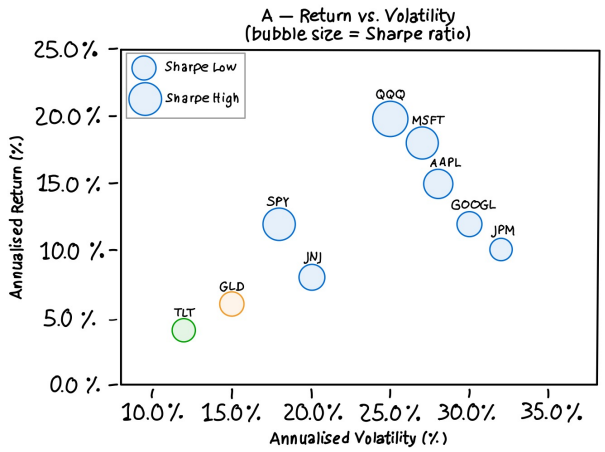
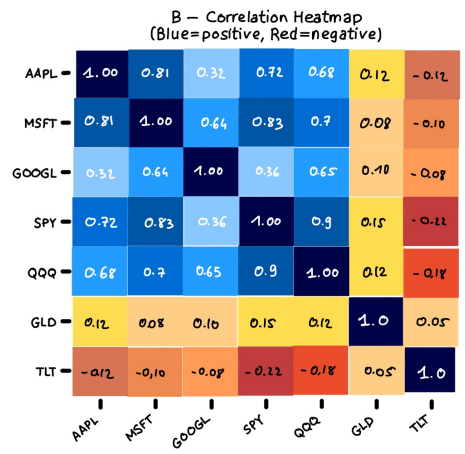
1. Visualization Sketches

Wireframe sketches for the six core visualizations across three tabs (A–C: Asset Explorer, D: Risk Studio, E–F: Portfolio Builder).

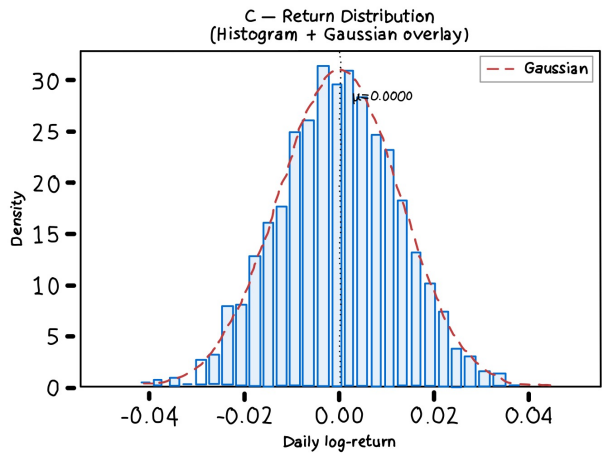
A — Return vs. Volatility



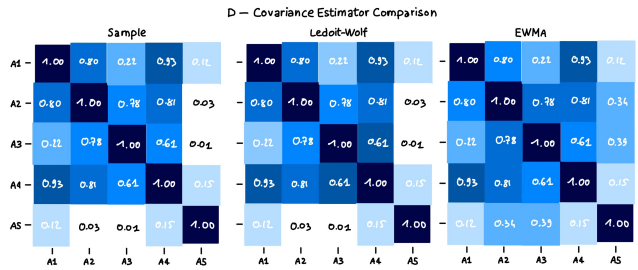
B — Correlation Heatmap



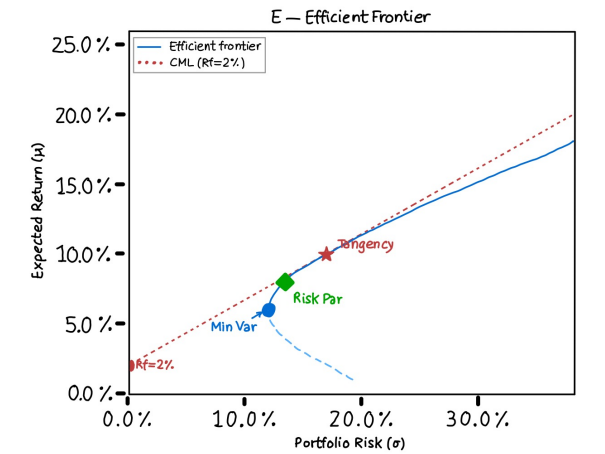
C — Return Distribution



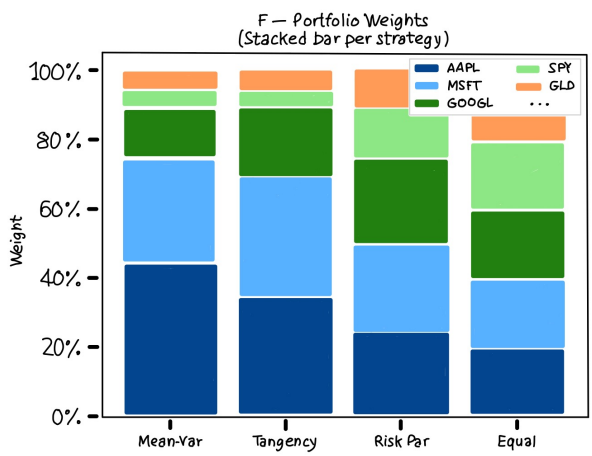
D — Covariance Estimator Comparison



E — Efficient Frontier



F — Portfolio Weights



2. Tools & Lecture Mapping

Visualization	Tools / Libraries
A. Return vs. Volatility scatter	D3.js (scales, axes, circles, tooltips), Sharpe ratio computation, Capital Market Line (CML) overlay
B. Correlation / Covariance heatmaps	D3.js (grid rendering), chroma.js (diverging color scales), covariance-to-correlation transformation
C. Return distribution histogram	D3.js (d3.bin, linear scales), Gaussian overlay, skewness and kurtosis computation
D. Covariance models comparison	Vanilla JS + math.js, Sample / Ledoit-Wolf / EWMA covariance estimators, eigenvalue decomposition
E. Efficient frontier	D3.js (scatter plot), portfolio optimisation (min variance, tangency, risk parity, mean-variance), gradient descent solver
F. Portfolio composition	D3.js stacked bar, portfolio weights visualization, risk contribution decomposition

Relevant lectures

- Interactive data visualization with D3.js (Lec 4–5)
- Visual encoding, perception and color scales (Lec 6)
- Statistical distributions and risk metrics (Lec 6)
- Portfolio theory and optimization methods (Lec 5–7)

Additional: HTML/CSS/JS (Lec 1-2-3: Web Dev and Javascript part 1 & 2); GitHub Pages for hosting (Lec 7: Deployment); storytelling principles (Lec 12) for guiding progressive disclosure across tabs.

3. Core Visualization – Minimal Viable Product (MVP)

P1 – Data pipeline (JS):
Parse the static CSV with `d3.csv`, align dates, compute daily log-returns, and derive annualised statistics (mean, volatility, Sharpe, skewness, kurtosis) entirely client-side. No backend required.

P2 – Asset Explorer tab:
KPI summary row (number of assets, observation count, date range, best Sharpe, best asset). Interactive scatter plot of annualised return vs. volatility (bubble size = Sharpe). Correlation heatmap with diverging colour scale. Return distribution histogram with Gaussian overlay. Summary statistics table.

P3 – Risk Estimation Studio tab:
Compute and display three covariance matrices — Sample, Ledoit-Wolf (shrinkage), EWMA — as side-by-side heatmaps. Provide a difference-toggle to highlight how estimators diverge, especially on off-diagonal terms.

P4 – Portfolio Builder tab:
Optimisation engine for four strategies: Minimum Variance, Tangency (max Sharpe), Risk Parity, and Mean-Variance with a target-return slider. Display portfolio weights as horizontal stacked bars. Plot the efficient frontier curve with labelled special points (Min Var, Tangency, user portfolio). Draw the Capital Market Line from the risk-free rate through the tangency portfolio.

P5 – Linked interactions:
Asset chip selector and time-window buttons (1Y / 3Y / 5Y / 10Y) update all tabs simultaneously. Hover tooltips on every chart element. Smooth D3 transitions when data changes.

4. Extra Ideas (Enhancement Layer – Can Be Dropped)

E1 – Cumulative return line chart:
Zoomable time series showing growth of $1 invested per asset, with a brushable date-range selector for sub-period analysis.

E2 – Rolling statistics panel:
Rolling Sharpe ratio, rolling volatility, and rolling pairwise correlation with configurable window (30 / 60 / 90 / 252 days).

E3 – Monte Carlo feasible set:
Simulate N random portfolios (uniform Dirichlet weights) and scatter them on the risk-return plane to visualise the full feasible region before overlaying the efficient frontier.

E4 – Custom weight playground:
Draggable sliders for each asset weight (summing to 100 %) that place the user's custom portfolio on the frontier plot in real time.